Aft-Fan Inflow Induction and Wing Separation During Tail-Sitter Transition

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A tail-sitter with an aft-mounted ducted fan spends its transition at incidences where the wing is separated, and the fan sits close enough behind the wing to change that. The mechanism is not the one a blown-wing study would model: the duct leading edge lies a fifth of a chord *aft* of the wing trailing edge, so its slipstream never reaches the wing. What reaches the wing is the fan's *upstream induction* — the accelerating field ahead of a loaded disk — which is a favourable streamwise pressure gradient and therefore delays separation. This paper models that induction as the semi-infinite cylindrical vortex sheet equivalent to a uniformly loaded actuator disk, verified against the closed-form axial solution and against momentum theory, and applies it as an external field to a coupled wing–body panel solve whose separation behaviour is established independently. On the demonstrator airframe the fan is found to buy $\Delta\alpha \approx 0.8°$ of incidence at the transition point, measured as the shift of the separation-location curve rather than as the crossing of a single threshold — a distinction that matters, since a threshold evaluated on a two-degree grid reports no effect at

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all from data containing this one. The benefit is concentrated inboard, over precisely the span station that separates first, and reverses sign outboard where the wing leaves the fan's streamtube and meets its return flow.

Nomenclature

A	disk area, m^2
b	wing span, m
c_n	chord in the leading-edge-normal plane, m
C_L	lift coefficient
R	disk radius, m
s	axial distance from the disk plane, m
T	fan thrust, N
u_x	axial induced velocity, m/s
v_h	hover induced velocity, $\sqrt{T/(2\rho A)}$, m/s
v_i	induced velocity at the disk plane, m/s
V_∞	freestream speed, m/s
x_{sep}	chordwise separation station, fraction of c_n
α	angle of attack, deg
γ_t	cylindrical vortex sheet strength, m/s
μ	velocity ratio, V_∞/v_h
ρ	density, kg/m^3

I. Introduction

Transition is the part of a tail-sitter's envelope that sizes its wing and its control authority,[1,2] and it is the part a potential-flow method is least able to speak to, because the wing is separated over much of it. The companion study of this airframe[3] establishes where separation begins and how far forward it moves with incidence, marching an integral boundary layer from the located attachment point.[4,5] This paper asks what the propulsion system does about it.

The question is usually posed for a *tractor* installation, where the propeller slipstream washes the wing and raises its dynamic pressure — a blown wing, studied exhaustively for both conventional and wingtip-mounted layouts.[6,7] The airframe here is the other arrangement, and the difference is not a detail. Its ducted fan — a configuration whose static performance and lip-suction thrust split are classical[8,9] and whose small-scale behaviour is well characterized[10] — sits at the tail: the duct leading edge is 0.21 chord *behind* the wing trailing edge, so the slipstream leaves downstream and never touches the wing. A model that represents only the wake therefore reports exactly zero interaction, which is a statement about the model rather than about the aircraft.

What does reach the wing is the induction *ahead* of the disk. A loaded actuator disk accelerates the flow approaching it, from zero far upstream to v_i at the disk plane;[11,12] a wing sitting in that field experiences a favourable streamwise pressure gradient, and a favourable gradient delays separation.[13] The effect is the same in sign as a blown wing's and entirely different in mechanism. Its closest studied analogue is the upstream blockage of wind-turbine rotors, where the same vortex-cylinder model is used to quantify how far ahead of a rotor the flow is disturbed;[14] the present application reverses the question, asking what that disturbance does to a lifting surface placed inside it.

Figure effig:configuration shows the configuration and the field in question. Two features make it sharp rather than academic. First, the induced velocity is not a small correction: with the annular disk area of the demonstrator, v_h runs 14–24 m/s against a 25 m/s cruise, so the induction is comparable to the flight speed throughout transition and largest exactly where the airspeed is lowest. Second, the geometry aligns: the duct outer radius is 0.21 semi-span, and the companion study places the wing's most advanced separation at $|2y/b| \approx 0.15$–0.23. The part of the wing that sheds first is the part the fan protects.

II. The induction model

A. Why the wake model cannot serve

The solver's existing slipstream model reconstructs a rotor's wake from annular momentum theory[12] and returns zero induced velocity at any point ahead of the disk plane. That is a correct economy for a surface lying in the wake, and it is the wrong tool here by construction. Applied to a wing upstream of the disk it returns zero, and the zero is the model's rather than the flow's.

B. The disk as a vortex cylinder

A uniformly loaded actuator disk is exactly equivalent to a semi-infinite cylindrical vortex sheet.[15] The equivalence is worth deriving rather than asserting, because everything downstream of it — including the sign of the effect this paper reports — follows from the derivation.

Across a disk of radius R carrying uniform loading, the pressure jump Δp is constant over the disk face and the bound circulation on a blade element is therefore independent of

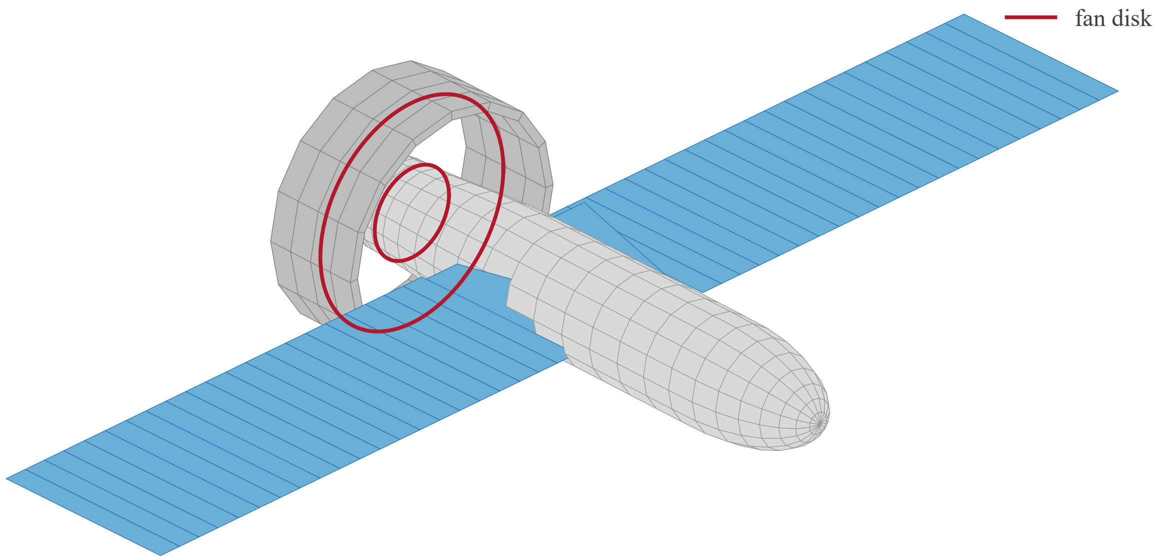

Figure 1. The coupled solve: the wing lattice, the source-panelled fuselage and duct, and the fan disk, all carried in one system. The fan sits behind the wing and inboard of its mid-span — the two geometric facts the rest of this paper turns on.

radius. Trailing vorticity is shed only where the bound circulation *changes*, so for uniform loading it is shed only at the edge $r = R$, and the shed sheet is convected downstream to form a cylinder. Its strength follows from the velocity jump across the sheet: far downstream the flow inside the streamtube is faster than outside by the fully developed wake increment, so

$$\gamma_t \;=\; [u_x]^{\text{in}}_{\text{out}} \;=\; 2v_i, \tag{1}$$

per unit axial length, with the vorticity purely azimuthal. This is the hydrodynamic analogue of a solenoid, and the axial field follows the same way.

The axis solution. Take the disk plane at $s = 0$ with the cylinder occupying $s \in [0, \infty)$. A ring element of the sheet at axial station s' carries circulation $\mathrm{d}\Gamma = \gamma_t\,\mathrm{d}s'$, and a circular vortex filament of radius R and circulation $\mathrm{d}\Gamma$ induces, on its own axis at a distance $\xi = s - s'$,

$$\mathrm{d}u_x = \frac{\mathrm{d}\Gamma}{2}\,\frac{R^2}{(R^2 + \xi^2)^{3/2}}. \tag{2}$$

Integrating Eq. (2) over the whole semi-infinite cylinder,

$$u_x(s) = \frac{\gamma_t R^2}{2} \int_0^\infty \frac{\mathrm{d}s'}{[R^2 + (s - s')^2]^{3/2}} = \frac{\gamma_t}{2} \left[1 + \frac{s}{\sqrt{s^2 + R^2}} \right], \tag{3}$$

which is the closed form the model is verified against. Its three limits are the three statements of momentum theory, recovered from the vortex system rather than assumed:

$$u_x(-\infty) = 0, \qquad u_x(0) = \frac{\gamma_t}{2} = v_i, \qquad u_x(+\infty) = \gamma_t = 2v_i. \tag{4}$$

The middle limit is the classical result that half the ultimate induction is present at the disk, and here it is a consequence of the cylinder being *semi*-infinite: the disk plane sees exactly half the sheet.

THE UPSTREAM FIELD, WHICH IS WHAT THIS PAPER NEEDS. Setting $s < 0$ in Eq. (3) gives the induction ahead of the disk, and expanding for $|s| \gg R$,

$$u_x(s) \;=\; \frac{\gamma_t}{2} \left[1 - \frac{|s|}{\sqrt{s^2 + R^2}} \right] \;\sim\; \frac{v_i R^2}{2s^2} \qquad (s \to -\infty), \tag{5}$$

so the upstream induction decays as the inverse square of distance — the signature of a semi-infinite vortex cylinder viewed from far upstream, where it appears as a point sink of strength $\pi R^2 \gamma_t$. The decay is fast, which is why the effect reported here is a near-field one and why the wing's proximity to the duct is essential rather than incidental. At one radius upstream Eq. (3) gives

$$u_x(-R) = \frac{\gamma_t}{2} \left(1 - \frac{1}{\sqrt{2}} \right) = 0.293\, v_i, \tag{6}$$

the number the whole interaction argument turns on, and the one Fig. 3 marks.

Off the axis the sheet is discretized into vortex rings, each polygonized into straight filaments and evaluated with the same Biot–Savart kernel the lattice uses for its own wake. Ring spacing grows geometrically downstream, since the first radius or two sets essentially the entire upstream answer while the far tail varies slowly, and that spacing keys to each cylinder's own radius rather than the disk's outer one.

THE ANNULUS. A ducted fan around a tail boom is an annulus, not a disk, and the same argument gives it directly. Uniform loading between R_h and R sheds trailing vorticity only where the bound circulation changes, which is now at *both* edges: $+\gamma_t$ at the outer radius and $-\gamma_t$ at the inner. The field is the superposition of two semi-infinite cylinders of opposite sign, so on the axis

$$u_x(s) = \frac{\gamma_t}{2}\left[\frac{s}{\sqrt{s^2+R^2}} - \frac{s}{\sqrt{s^2+R_h^2}}\right], \tag{7}$$

the constant terms of Eq. (3) having cancelled. Three consequences are worth reading off, because they are physical rather than numerical. The induction vanishes on the axis *at the disk plane* and at both infinities, since R and R_h enter symmetrically in those limits. Upstream it is positive but weaker than the equivalent solid disk's, the bore shielding the axis. And downstream inside the bore it is *negative*: for $s > 0$ the inner term dominates because $R_h < R$, so the flow on the axis behind an annular disk is retarded. That is the centerbody wake, a real feature of annular jets, and the model produces it without being told.

Swirl is deliberately absent from Eqs. (3) and (7). The root vortex and the wake's axial vorticity lie downstream of the disk and contribute nothing ahead of it, which is the region this model exists to serve.

C. Verification

The discretized sheet is checked against Eq. (3) at stations from four radii upstream to ten downstream, holding to 2% of v_i, with the transverse components vanishing on the axis by symmetry and the error falling under refinement. The annulus is checked against its own closed form, which pins the inner sheet's sign and strength, including the reversed induction inside the bore. The momentum inversion $T = 2\rho A v_i(V_\infty + v_i)$ round-trips, and forward speed reduces v_i at fixed thrust.

One further check is the one that magnitude comparisons cannot make: the induction must point *downstream* for a thrusting disk, both in the jet and ahead of it, and must reverse when the disk axis reverses. Inverting the ring circulation sense would turn the upstream acceleration this paper rests on into a deceleration while leaving every magnitude correct.

III. Application to the demonstrator

The induction is applied as an external perturbation field to the coupled wing–body–duct solve of the companion study,[3] and the separation march is run with the fan on and off at otherwise identical conditions. The coupling is presently *one-way*: the airframe sees the fan, the fan does not see the airframe, so every result below is a lower bound on the interaction.

Table 1. Chordwise rise in axial velocity across the wing, as a percentage of freestream, at four points along a transition. The velocity ratio $\mu = V_\infty/v_h$ runs from near-hover to cruise.

V_∞	T	v_i	μ	$\Delta u_x/V_\infty$ [%] at $2y/b =$			
[m/s]	[N]	[m/s]		0.10	0.20	0.30	0.60
5	25	17.1	0.26	65.7	40.4	13.9	-1.7
12	25	14.3	0.62	23.0	14.1	4.8	-0.6
18	20	10.6	1.04	11.3	6.9	2.4	-0.3
25	12	5.9	1.86	4.5	2.8	1.0	-0.1

A. Magnitude and distribution

The quantity that matters is not the induction at a point but its variation *along the chord*, because that variation is the pressure gradient. The wing trailing edge sits roughly three times closer to the disk than its leading edge, so the induction rises substantially across the chord: at the transition point $V_\infty = 12$ m/s, $T = 25$ N, the rise is 23% of freestream at $2y/b = 0.10$, 14.5% at 0.20 and 5% at 0.30. It is a favourable gradient, and it is strongest at the trailing edge, which is where turbulent separation begins and from which it marches forward.

Two features of the spanwise distribution deserve emphasis, and Fig. 4 and Table 1 carry both. It falls by a factor of four between $2y/b = 0.10$ and 0.30, concentrating the benefit inboard — over the stations that separate first.

And beyond $2y/b \approx 0.4$ it *reverses.* That sign change is not numerical, and Eq. (3) explains it: a semi-infinite vortex cylinder induces flow along its axis *inside* the streamtube and against it outside, since the total flux through any plane must be conserved and the disk adds none. A wing spanning both regions is therefore helped inboard and penalized outboard. The penalty is under 2% of freestream, an order of magnitude smaller than the inboard benefit, and it is a real property of the configuration that exists in the answer only because the upstream field is modelled at all.

B. How much incidence the fan buys

Measuring the benefit requires care, and the obvious measurement fails. Defining separation onset as a threshold crossing — the first α at which any station separates forward of a fixed chordwise station — and evaluating it on the two-degree grid used for the coefficient tables returns *no change at all* between fan on and fan off. That result is an artifact: a shift smaller than the grid spacing moves the crossing within an interval without moving it across one, so the reported onset is quantized to the grid and any sub-degree shift is invisible by construction.

Table 2. Incidence bought by the fan, as the horizontal shift of the separation curve at several criterion levels. $V_\infty = 12$ m/s, $T = 25$ N.

x_{sep}/c_n	α off [deg]	α on [deg]	$\Delta\alpha$ [deg]
0.90	4.52	5.05	+0.52
0.85	6.18	6.91	+0.73
0.80	8.36	9.17	+0.81
0.75	9.96	10.75	+0.79
0.70	11.09	11.95	+0.86
0.60	13.04	13.89	+0.85
0.55	14.08	14.88	+0.80
		mean, 0.85 to 0.55	+0.81

The appropriate measurement is the horizontal shift of the separation curve $x_{\text{sep}}(\alpha)$, evaluated at several criterion levels — Fig. 5 and Table 2. Measured that way on a 0.5° grid, the fan buys $\Delta\alpha = +0.81°$ averaged over criteria from $x_{\text{sep}} = 0.85$ to 0.55, with individual values between +0.73° and +0.86°.

The consistency across criterion levels is what makes this a property of the flow rather than of the definition: a spurious shift arising from the choice of threshold would vary with it. The shift is smaller near onset, +0.52° at $x_{\text{sep}} = 0.90$, and the reason is physical rather than numerical — there the separation point still lies at the trailing edge, where the boundary layer is thinnest and least sensitive to a gradient, so the same induction moves it least.

The benefit scales as expected with the velocity ratio: at cruise speed the same comparison returns roughly a third of the transition value, consistent with the induction scaling as v_i/V_∞.

IV. Limitations

The coupling is one-way, so the reported shift is a lower bound. The induction model assumes uniform disk loading; real loading tapers at both edges, softening the sheet into a band that superposed cylinders at intermediate radii would represent. The fuselage boundary layer that an aft fan genuinely ingests is not modelled, and it acts to reduce the benefit. The analysis is quasi-steady, while transition is a manoeuvre rather than a sequence of trim points. And the separation criterion inherits the companion study's caveat: bubble bursting[16] is not modelled, so the separation boundary is an upper bound on usable incidence rather than a predicted stall.

V. Conclusions

[Draft: conclusions pending the two-way coupling and the transition operating line. See the repository README.]

Acknowledgments

This work was supported by the Istanbul Technical University Scientific Research Projects Coordination Unit (İTÜ BAP) under project MGA-2026-48282, "JETTAIL: Design, Control, and Autonomous Mission Capability Development of a Jet-Flow-Vectored EDF-Based Tail-Sitter VTOL UAV."

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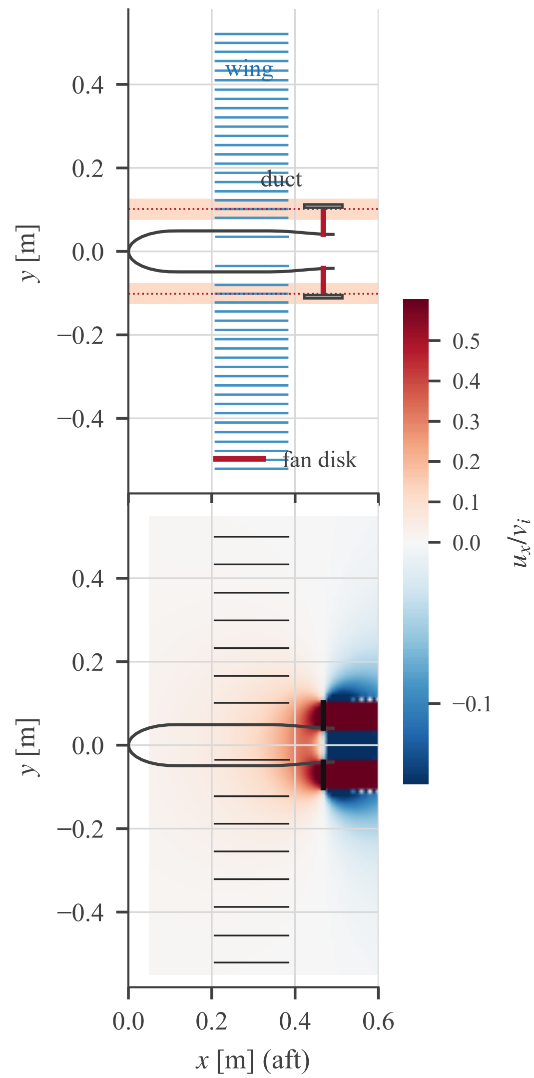

Figure 2. The coupled configuration being solved, in section. Top: planform, showing the wing lattice (44 strips, one Weissinger row each), the fuselage outline, the duct annulus and the fan disk edge-on. The shaded bands mark the span stations that separate first; the dotted lines are the fan tip radius, and their coincidence is the geometric claim of Sec. efsec:model. Bottom: the same geometry with the axial induction u_x/v_i the fan puts on it, at $V_\infty = 12$ m/s, $T = 25$ N. The wing lies in the accelerating (red) region ahead of the disk; outside the streamtube the induction reverses.

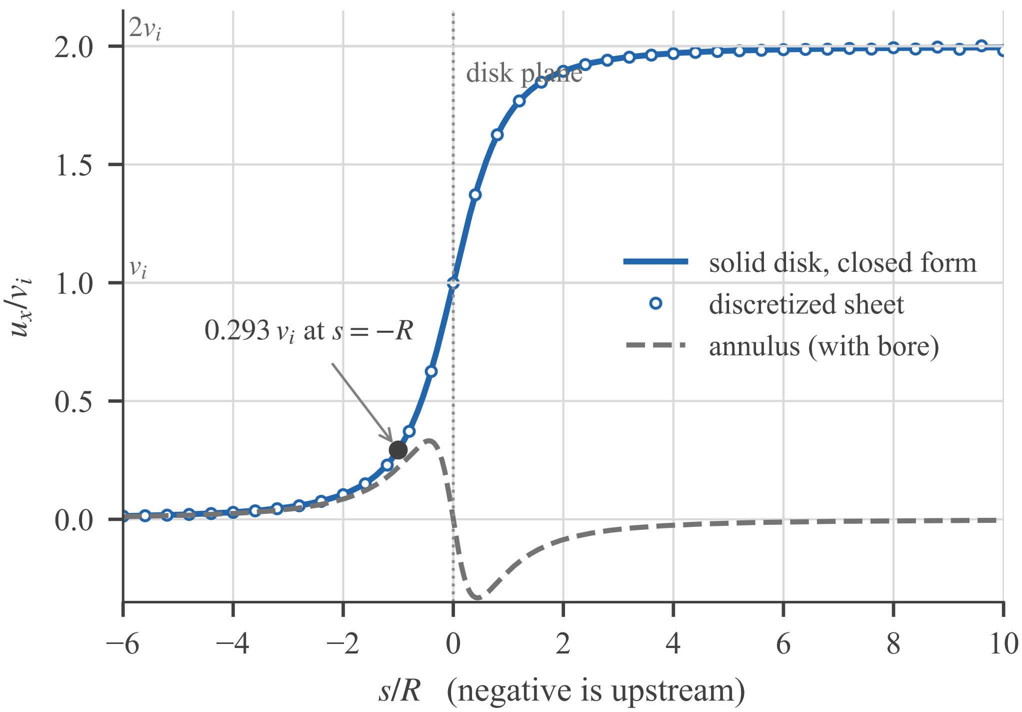

Figure 3. Axial induction on the disk axis. The discretized sheet (circles) against the closed form of Eq. (3) (solid), with the annulus (dashed) showing the reversal inside the bore that Eq. (7) predicts. Negative s is upstream — the half of the field a wake-only model does not carry, and the half this paper depends on.

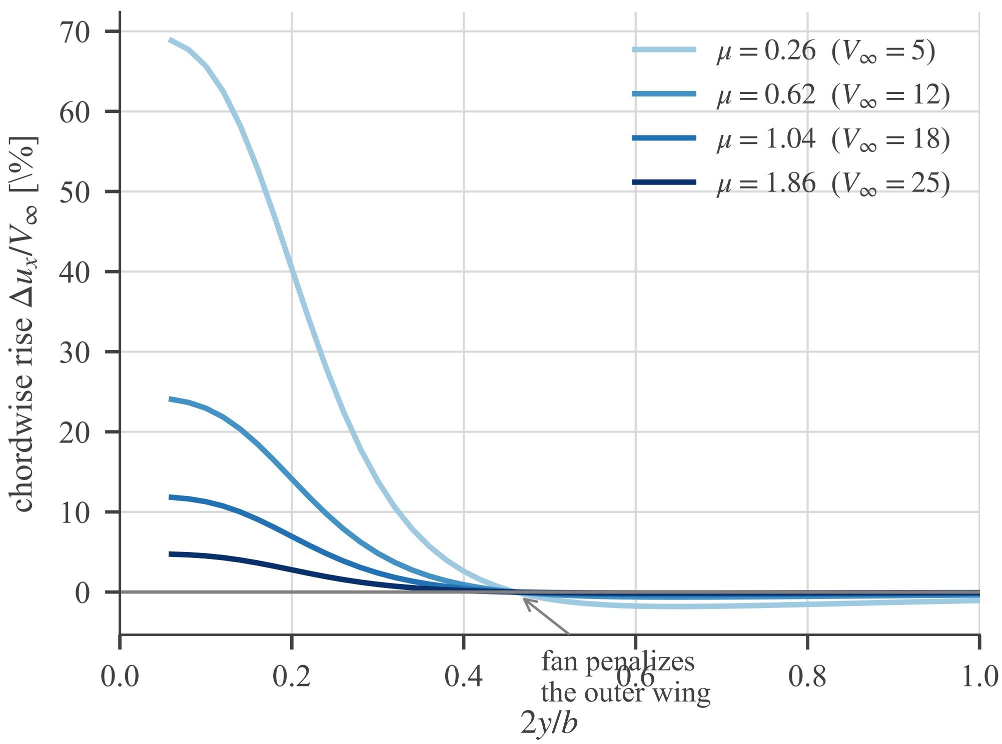

Figure 4. The chordwise rise across the span. The benefit is concentrated inboard — over the stations that separate first — and reverses beyond $2y/b \approx 0.4$, where the wing lies outside the fan's streamtube and meets its return flow.

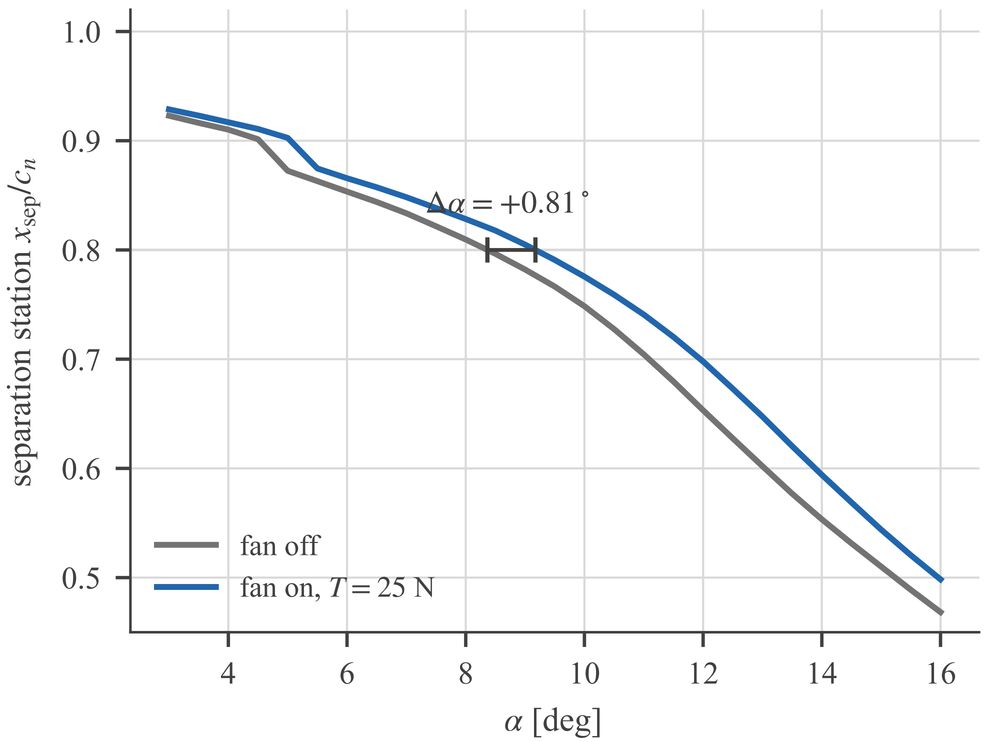

Figure 5. Separation station against incidence, fan off and on, at $V_\infty = 12$ m/s. The deliverable is the *horizontal* distance between the curves, marked at one criterion level; measuring instead where either curve crosses a fixed threshold, on a two-degree grid, returns no effect at all.